

[EECS 304] Control I

Sheet (4) Transient Response

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Overview

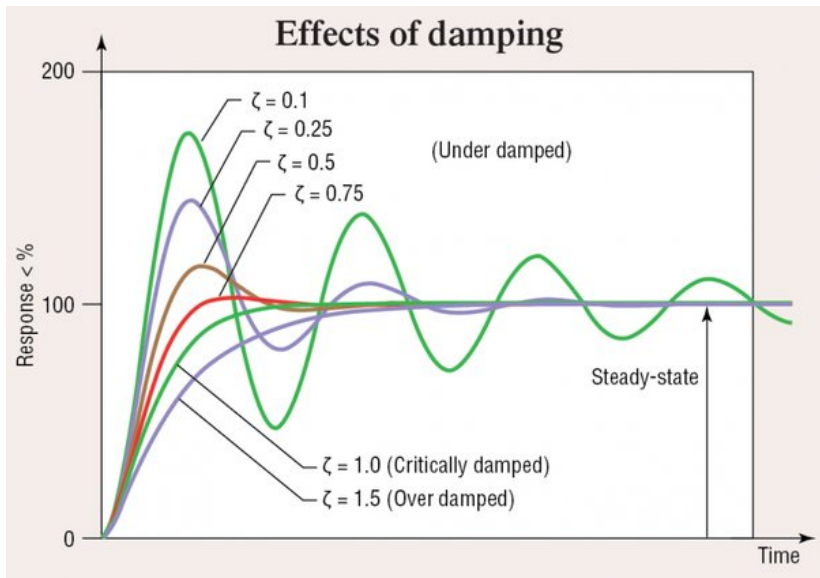
Maximum Overshoot	M_p	$e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$
Settling Time	t_s	$\frac{4}{\zeta\omega_n}$ (2% criterion), $\frac{3}{\zeta\omega_n}$ (5% criterion)
Rise Time	t_r	$\frac{\pi - \cos^{-1}\zeta}{\omega_n\sqrt{1-\zeta^2}}$
Peak Time	t_p	$\frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$
2 nd Degree Equation		$S^2 + 2\zeta\omega_n S + \omega_n^2 = 0$
3 rd Degree Equation		$(S + \alpha)(S^2 + 2\zeta\omega_n S + \omega_n^2) = 0$
Characteristic Equation		$1 + G(S)H(S) = 0$

- If you are solving a 3rd degree system and have only three equations (4 unknowns, 3 equations), assume $\alpha = 5\zeta\omega_n$ as your fourth equation.
- To get ζ from a given M_p , either solve a numerical method (SOLVE on the calculator) or use the rule:

$$\zeta = \frac{1}{\sqrt{1 + \left(\frac{\pi}{\ln M_p}\right)^2}}$$

- Pole placement: $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

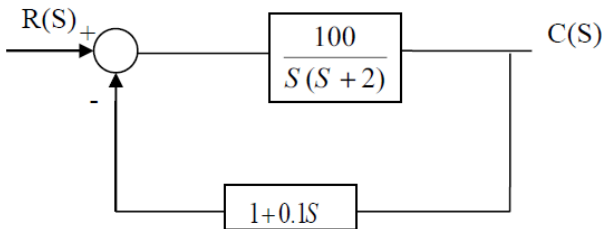
Damped Responses



Problem 1

Problem 1

The block diagram of a feedback control system is shown in the following figure. Determine the values of ζ , ω_n , M_p and t_s .



Problem 1

$$\text{Closed loop T.F.} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\text{Closed loop T.F.} = \frac{100}{s(s+2) + 100 + 10s} = \frac{100}{s^2 + 12s + 100}$$

From second order system:

$$c/ch \text{ eq.} : s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Compare:

$$s^2 + 12s + 100 = 0$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore \omega_n^2 = 100 \rightarrow \omega_n = 10$$

$$\therefore 2\zeta\omega_n = 12 \rightarrow \zeta = 0.6$$

Problem 1

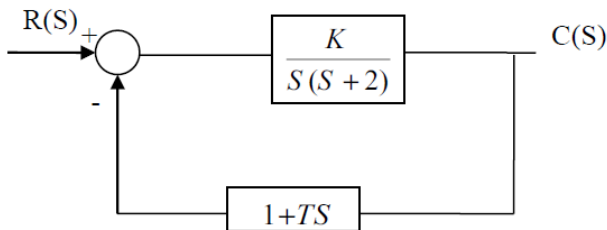
$$\therefore M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.09489 = 9.489\%$$

$$\therefore t_s = \frac{4}{\zeta\omega_n} = 0.6667\text{sec}$$

Problem 2

Problem 2

Referring to the system shown in figure, determine the values of K and T such that the system has a $\zeta = 0.7$, and $\omega_n = 4$ rad/sec



Problem 2

$$\text{Closed loop T.F.} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\text{Closed loop T.F.} = \frac{K}{s(s+2) + K + KTs} = \frac{K}{s^2 + (2 + KT)s + K}$$

From second order system:

$$c/ch \text{ eq. : } s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore s^2 + 5.6s + 16 = 0$$

Compare:

$$s^2 + (2 + KT)s + K = 0$$

$$s^2 + 5.6s + 16 = 0$$

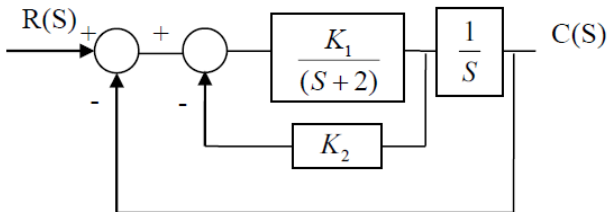
$$\therefore K = 16$$

$$\therefore T = \frac{5.6 - 2}{K} = 0.225$$

Problem 3

Problem 3

Referring to the system shown in figure, determine the values of K_1 and K_2 such that the system has a maximum overshoot in unit step response is 25% and the peak time is 2 sec.



Problem 3

$$M_p = 0.25 \rightarrow \zeta = 0.4$$

$$t_p = 2 \text{ sec} \rightarrow t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \rightarrow \omega_n = 1.715 \text{ rad/sec}$$

$$c/ch \text{ equation} : 1 + GH = 0$$

$$G = G_1 G_2$$

$$G_1 = \frac{K_1}{s + 2 + K_1 K_2} \quad G_2 = \frac{1}{s}$$

$$\therefore G = \frac{K_1}{s^2 + (2 + K_1 K_2)s}$$

$$\therefore c/ch \text{ equation} \rightarrow 1 + \frac{K_1}{s^2 + (2 + K_1 K_2)s} = 0$$

Problem 3

$$\therefore s^2 + (2 + K_1 K_2)s + K_1 = 0$$

$$\therefore s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore K_1 = \omega_n^2 = 2.94$$

$$\therefore 2 + K_1 K_2 = 2\zeta\omega_n$$

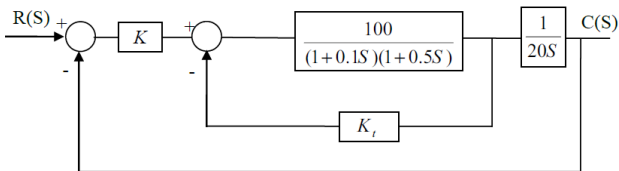
$$\therefore K_2 = -0.2316$$

Problem 4

Problem 4

For the control system shown in the figure:

- 1 Determine the values of K and K_t such that the output has a maximum overshoot 4.3% and the rise time is approximately 2 sec.
- 2 With the values of K and K_t obtained, find the steady state error when the input is a unit ramp function.



Problem 4

1

$$M_p = 0.043 \rightarrow \zeta = 0.7075$$

$$t_r = 2 \text{ sec} \rightarrow t_r = \frac{\pi - \cos^{-1}\zeta}{\omega_n \sqrt{1 - \zeta^2}} \rightarrow \omega_n = 1.668 \text{ rad/sec}$$

$$c/ch \text{ equation} : 1 + GH = 0$$

$$G = G_1 G_2 G_3$$

$$G_1 = K \quad G_2 = \frac{100}{0.05s^2 + 0.6s + (1 + 100K_t)} \quad G_3 = \frac{1}{20s}$$

$$\therefore G = \frac{100K}{s^3 + 12s^2 + (20 + 2000K_t)s}$$

$$\therefore c/ch \text{ equation} \rightarrow 1 + \frac{100K}{s^3 + 12s^2 + (20 + 2000K_t)s} = 0$$

Problem 4

$$\therefore s^3 + 12s^2 + (20 + 2000K_t)s + 100K = 0$$

$$\therefore (s + \alpha)(s^2 + 2\zeta\omega_n s + \omega_n^2) = 0$$

$$\therefore \alpha + 2\zeta\omega_n = 12$$

$$\therefore \omega_n^2 + 2\zeta\omega_n\alpha = 20 + 2000K_t$$

$$\therefore \alpha\omega_n^2 = 100K$$

$$\therefore \alpha = 9.646$$

$$\therefore K_t = 2.69 * 10^{-3}$$

$$\therefore K = 0.2658$$

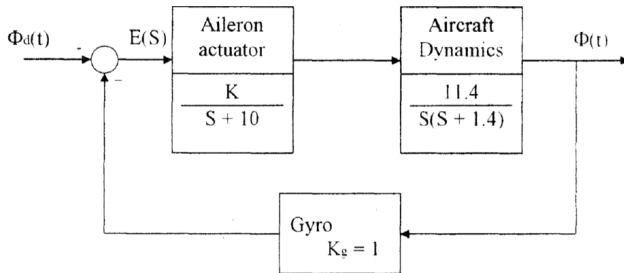
$$\textcircled{2} \quad e_{ss}|_{unit \text{ ramp}} = \frac{1}{K_v}$$

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s) = 1.0473 \rightarrow e_{ss} = 0.95484$$

Problem 5

Problem 5

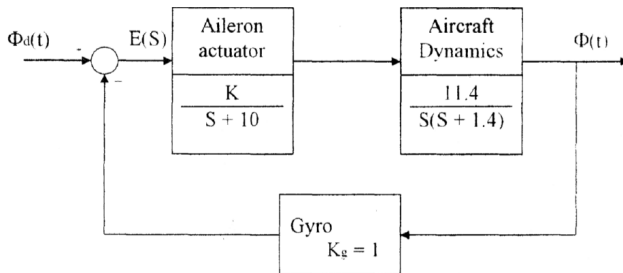
The roll control autopilot of a jet fighter is shown in the below figure. The goal is to select a suitable K so that the response to a unit step command $\phi_d(t)$ will provide a response $\phi(t)$ that is a fast response and has an overshoot of less than or equal 9.5%:



Problem 5

Problem 5

- 1 Using the concept of dominant poles, find a suitable value of K that will achieve the desired transient response. Predict the transient response of the system (i.e. get t_r , t_p and t_s).
- 2 Find the static error coefficients of the system. Evaluate the value of K that gives minimum steady state error for a unit ramp input.



Problem 5

1

$$M_p = 0.095 \rightarrow \zeta = 0.5998$$

$$c/ch \text{ equation} \rightarrow s^3 + 11.4s^2 + 14s + 11.4K = 0$$

$$\therefore (s + \alpha)(s^2 + 2\zeta\omega_n s + \omega_n^2) = 0$$

I will have 3 equations in 3 unknowns:

$$\therefore \alpha + 2\zeta\omega_n = 11.4$$

$$\therefore \omega_n^2 + 2\zeta\omega_n\alpha = 14$$

$$\therefore \alpha\omega_n^2 = 11.4K$$

$$\therefore K = 0.9973$$

$$\therefore \alpha = 10.13$$

$$\therefore \omega_n = 1.059 \text{ rad/sec}$$

Problem 5

$$\therefore t_r = 2.612 \text{ sec}$$

$$\therefore t_p = 3.706 \text{ sec}$$

$$\therefore t_s = 6.292 \text{ sec}$$

② to get min. $e_{ss} \rightarrow$ check stability

$$s^3 + 11.4s^2 + 14s + 11.4K = 0$$

For 3rd order system $\rightarrow a_2a_1 > a_3a_0$

$$\therefore 11.4 * 14 > 11.4K$$

$$\therefore K < 14$$

$$\therefore K_{max} = 14$$

$$\text{For unit ramp} \rightarrow e_{ss} = \frac{1}{K_v} = \frac{1}{0.8 * K_{max}} = \frac{1}{0.8 * 14} = 0.089$$

Thank you!

Questions?

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